RLHF Step-by-Step Experiment for Korean Model Alignment

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Abstract

This study presents an experimental analysis of Reinforcement Learning from Human Feedback (RLHF), a key technique for addressing the alignment problem in Large Language Models (LLMs). RLHF is a core alignment technology used to make models accurately and helpfully follow user intentions, and it has been employed in models such as OpenAI's ChatGPT (GPT-3.5, GPT-4), Google's Gemini, Anthropic's Claude, and Meta's Llama 2/3. However, for NLP researchers, merely observing existing research findings presents a challenge for practical learning. Especially for utilizing the technology in a different linguistic context like Korean, a direct, hands-on experience of its effects is crucial. Against this backdrop, this study aims to directly verify the effectiveness of RLHF by applying its pipeline to a Korean language model and analyzing the impact of dataset quality on the final model's performance. Specifically, we sought to observe the performance changes at each stage of the RLHF pipeline on a Korean model and to further deepen our understanding by varying the data quality. As a methodology, we implemented the three stages of RLHF—Supervised Fine-tuning (SFT), Reward Modeling (RM), and Reinforcement Learning with Proximal Policy Optimization (PPO)—using SKT's KoGPT-2 as the base model. An experiment using the dataset from the public KoChatGPT project was set as the baseline. For comparison, we conducted an additional experiment with an enhanced dataset, which we created by improving the quality using GPT-4o mini. To facilitate a multi-faceted evaluation of model performance, we developed a custom evaluation framework that includes the quantitative metric BERTScore, as well as qualitative measures such as usability and accuracy, using an LLM as an evaluator. The results showed that the PPO model trained on the new dataset (PPO_NewData) achieved the highest performance across all evaluation metrics compared to the base model (KoGPT-2) and all other experimental models. Performance improvements were particularly notable in complex tasks requiring subtle nuances, such as role-playing or style transfer. In conclusion, we quantitatively confirmed the effectiveness of the RLHF pipeline in enhancing the instruction-following capabilities of a Korean model and identified the mechanisms through which dataset quality influences model performance via RLHF.

1 Introduction

OpenAI's InstructGPT study ouyang2022training demonstrated that simply increasing a model's parameter count cannot solve the alignment problem, proposing fine-tuning with human feedback as an effective solution. This research introduced Reinforcement Learning from Human Feedback (RLHF)—optimizing a language model using human preferences as a reward signal—as a core technology for alignment, which serves as the theoretical foundation for our study.

There have been instances of RLHF implementation in the Korean context. The 'KoChatGPT' GitHub project ko2023kochatgpt by W. Ko, a researcher at an electronics and telecommunications research institute, provided a foundation for domestic researchers to experiment with RLHF by publicly releasing a Korean implementation of the pipeline based on SKT's KoGPT-2 model. As this project focused on sharing the methodology, it did not provide a quantitative analysis of the impact of each RLHF stage on model performance. This study aims to address this Research Gap.

Therefore, building upon the KoChatGPT project's implementation, this study sets the following objectives: (1) to quantitatively and qualitatively analyze the impact of each stage (SFT, PPO) on the instruction-following capabilities of SKT's KoGPT-2 model when applying the RLHF pipeline, and (2) to experimentally verify the effect of dataset quality on model performance during RLHF training. Through this, we seek to find empirical evidence for methods of aligning Korean LLMs.

The next chapter will delve deeper into the alignment problem in large language models, the RLHF methodology, and the preceding Korean-related studies that form the theoretical basis of this research.

2 Background and Related Works

To understand the experimental design of this study, this chapter sequentially summarizes the alignment problem in large language models, the core RLHF methodology to solve it, and the related Korean-language prior research that served as a direct starting point.

2.1 The Alignment Problem in LLMs

Large language models are trained to predict the next token from vast amounts of text data. This "language modeling objective" enables models to generate grammatically fluent and coherent text, but it is fundamentally different from the actual goal of "following user instructions usefully and safely." OpenAI's InstructGPT research defines this discrepancy between the training objective and the practical goal as the "misalignment" problem.

2.2 Reinforcement Learning from Human Feedback (RLHF)

RLHF is a methodology for aligning language models by directly utilizing human preference data. The RLHF pipeline proposed in the InstructGPT study consists of the following three steps:

2.2.1 Step 1: Supervised Fine-tuning (SFT)

A pre-trained language model (LM) is fine-tuned on a high-quality demonstration dataset of 'instruction-response' pairs written by human labelers. This step aims to equip the model with the basic ability to follow instructions.

2.2.2 Step 2: Reward Model (RM) Training

Multiple responses to the same prompt are collected, generated by the SFT model or various other model versions. Human labelers then evaluate these responses by ranking them according to their preference. This collected comparison dataset is used to train a 'Reward Model' that predicts which response a human would prefer. This reward model acts as a scorer, quantifying subjective human preferences.

2.2.3 Step 3: Reinforcement Learning with PPO

The reward model trained in Step 2 is used as the reward function for reinforcement learning. The SFT model from Step 1 serves as the **initial policy**, which is then optimized using the Proximal Policy Optimization (PPO) algorithm to generate responses that receive higher scores from the reward model. To maintain training stability, a penalty term using the KL divergence between the PPO policy and the initial SFT policy is added. This prevents the policy from straying too far from the initial distribution, which is an essential regularization term to prevent **'reward hacking'**—exploiting loopholes in the reward model to get high scores with nonsensical text—and to maintain linguistic coherence.

2.3 KoChatGPT Project

The 'KoChatGPT' GitHub project by Senior Researcher W. Ko is a significant prior work that directly motivated this study. It is one of the first public codebases to implement the entire three-stage RLHF pipeline in a Korean environment, based on SKT's KoGPT-2 model. The data construction process was as follows. First, a dataset of approximately 12,000 Korean questions was collected from various sources. For Step 1 SFT, responses to each question were automatically generated using the ChatGPT API. For Step 2 RM training, a comparison dataset was built by generating responses to the same questions using three models of different performance levels (ChatGPT-3, GPT-3 'text-davinci-003', GPT-3 'text-ada-001') and automatically assigning preference rankings based on model performance. This research provided a methodological foundation for researchers attempting RLHF in a Korean context. However, as previously mentioned, the project focused on the implementation itself, lacking a systematic evaluation and analysis of the performance of the models at each training stage. Our study starts from this point, aiming to reproduce the public pipeline and further analyze the effects of RLHF in-depth by introducing the variable of data quality. The specific research methods designed based on this background are described in the next chapter.

3 Methodology

This study designed its experimental methodology with two main points of investigation: "How does the RLHF pipeline improve the performance of a Korean language model at each stage?" and "How does the quality of the dataset enhance the effectiveness of RLHF?" This chapter describes the overall experimental flow, the base model and datasets used, and the evaluation system built to measure model performance.

3.1 Experimental Overview

The overall research flow consists of the following five stages:

1) **Evaluation System Construction:** Design of evaluation prompts and metrics to measure model performance from multiple quantitative and qualitative perspectives.
2) **Baseline Model Evaluation:** Measurement of the performance of the pure, unfine-tuned KoGPT-2 model to establish a baseline for performance improvement.
3) **Experiment 1 (Baseline):** Sequentially training SFT and PPO models using the public KoChatGPT dataset and evaluating the performance at each stage.
4) **Experiment 2 (Enhanced):** Training SFT and PPO models using a new, quality-improved dataset created with GPT-4o mini, and evaluating their performance.
5) **Results Analysis:** Comprehensively analyzing all experimental results to compare the effects of the RLHF stages and dataset quality on model performance.

3.2 Base Model and Datasets

3.2.1 Base Language Model

All experiments in this study used SK Telecom's publicly available Korean GPT-2 model, **skt/kogpt2-base-v2**, as the base model.

3.2.2 Baseline Dataset

Experiment 1 used the dataset released in the KoChatGPT GitHub project. This dataset was created by generating three responses for 12,000 Korean question prompts using three models of varying performance (ChatGPT-3, GPT-3 'text-davinci-003', GPT-3 'text-ada-001'). A comparison dataset for Reward Model (RM) training was then constructed by automatically assigning rankings based on the models' performance order (ChatGPT > davinci > ada).

3.2.3 Enhanced Dataset

In Experiment 2, we created an enhanced version of the baseline dataset. Among the models used for the original dataset, ChatGPT-3 was the best. We added responses generated by a superior model, GPT-4o mini, to the existing dataset and assigned it the highest feedback score. The Reward Model (RM) was thus trained to compare and rank a total of four responses.

3.3 RLHF Pipeline Implementation

3.3.1 Step 1: Supervised Fine-tuning (SFT)

Based on the KoGPT-2 model, fine-tuning was conducted for 3 epochs using the highest-quality question-response pairs from the baseline and enhanced datasets, respectively. This produced the `SFT` and `SFT_NewData` models.

3.3.2 Step 2: Reward Modeling (RM)

Separate Reward Models (RMs) were trained using the ranking data from the baseline dataset (comparing 3 responses) and the enhanced dataset (comparing 4 responses).

3.3.3 Step 3: Reinforcement Learning with PPO

PPO reinforcement learning was conducted using the SFT models (`SFT`, `SFT_NewData`) and their corresponding Reward Models (RMs). This produced the final models, `PPO` and `PPO_NewData`.

3.4 Evaluation System Construction

3.4.1 Evaluation Prompts

To comprehensively assess the model's diverse instruction-following capabilities, a total of 25 prompts were custom-made. The prompts were organized into five task-based categories: (1) Simple Q&A (5), (2) Summarization (5), (3) Creative Writing (5), (4) Classification and Reasoning (5), and (5) Role-playing and Style Transfer (5).

3.4.2 Evaluation Metrics

A dual evaluation system combining quantitative and qualitative assessments was established to evaluate the model-generated responses, as detailed in Table 1.

3.4.3 Implementation of the Evaluation Generator

A generator was developed to produce responses to the prompts using the five models saved from Experiments 1 and 2: KoGPT-2, SFT, SFT_NewData, PPO, PPO_NewData.

To enhance the response performance of the base model, the generator for KoGPT-2 was configured identically to the one released in the public KoChatGPT GitHub project.

For the other four models (KoGPT-2, SFT, SFT_NewData, PPO, PPO_NewData), the generators were configured identically to the one used for the SFT model in the KoChatGPT GitHub project. This was to ensure a fair and consistent performance comparison among the models.

The Overall LLM score (`All_LLM_score`) is calculated as a weighted average, with weights of 'Usability': 0.3, 'Accuracy': 0.3, 'Consistency': 0.2, and 'Completeness': 0.2. The results are presented in Chapter 5.

Table 1: Evaluation Metrics Framework

Metric Type	Description
Quantitative (BERTScore)	For 13 prompts with reference answers, the semantic similarity between the model's response and the reference answer is measured as an F1 score.
Qualitative (LLM-based)	An LLM was used as an evaluator. Each response was rated on a 1-10 scale across four criteria: **Usability**, **Accuracy**, **Consistency**, and **Completeness**.

4 Results

This chapter analyzes and compares the performance of the five models (KoGPT-2, SFT, SFT_NewData, PPO, PPO_NewData) using the designed evaluation system. The analysis begins with an overall performance comparison, followed by the effects of the RLHF stages, the impact of dataset quality, and a task-specific analysis.

4.1 Overall Performance Comparison

The results for the weighted average score from the LLM-based qualitative evaluation (`All_LLM_score`) for the five models are shown in Table 2.

Table 2: Overall LLM Score Comparison

Model	Overall LLM Score
KoGPT-2 (Base Model)	2.17
SFT (Baseline Data)	2.29
PPO (Baseline Data)	2.32
SFT_NewData (Enhanced Data)	2.33
PPO_NewData (Enhanced Data)	2.60

The analysis reveals that the `PPO_NewData` model achieved the highest overall score of 2.60, indicating a clear trend of gradual improvement in the model's instruction-following ability as it progresses through the RLHF stages with improved data quality.

4.2 Analysis of Performance Changes by RLHF Stage

We tracked the performance changes for each dataset type to analyze the impact of each RLHF stage.

- **Baseline Dataset:** Applying SFT to the base model KoGPT-2 (2.17) improved the score to 2.29, with a minor additional improvement to 2.32 through PPO. The improvement from the PPO stage was limited to 0.03 points.
- **Enhanced Dataset:** Applying SFT with the new data to KoGPT-2 (2.17) resulted in a score of 2.33 for the `SFT_NewData` model. Notably, the subsequent PPO

stage led to a large performance increase to 2.60. The performance gain at this PPO stage was 0.27 points, approximately 9 times greater than that observed with the baseline dataset (0.03), which we attribute to the higher quality comparison data improving the Reward Model (RM).

4.3 Performance Comparison by Dataset Quality

To directly compare the impact of dataset quality, we compared models at the same training stage.

- **SFT Stage Comparison:** The `SFT_NewData` model (2.33) outperformed the `SFT` model (2.29).
- **PPO Stage Comparison:** The effect was more pronounced at the PPO stage. The `PPO_NewData` model (2.60) showed a significant performance gap of 0.28 points over the baseline PPO model (2.32).

In both comparisons, models trained with the new dataset performed better, demonstrating that the reward signal from the enhanced dataset provides a clearer direction for policy optimization.

4.4 In-depth Task-specific Analysis

Table 3 shows the detailed scores by task type.

Table 3: Task-specific Performance Analysis

Category	Model	bertscore_f1	Usability	Accuracy	Consistency	Completeness	All_LLM_score
Overall	KoGPT-2	0.59	2.04	2.40	2.60	1.56	2.17
	SFT	0.61	2.12	2.48	2.64	1.76	2.29
	SFT_NewData	0.62	2.36	2.20	2.64	1.96	2.33
	PPO	0.61	2.24	2.28	2.80	1.92	2.32
	PPO_NewData	0.60	2.44	2.76	3.24	1.96	2.60
Simple Q&A	KoGPT-2	0.54	1.00	1.00	1.80	1.00	1.16
	SFT	0.58	2.20	1.80	2.40	1.80	2.04
	SFT_NewData	0.59	2.20	2.20	2.40	1.40	2.08
	PPO	0.57	2.20	1.60	3.20	1.40	2.06
	PPO_NewData	0.55	3.00	2.20	3.60	2.40	2.76
Summarization	KoGPT-2	0.67	2.20	3.00	3.20	1.60	2.52
	SFT	0.65	2.20	2.80	3.40	1.80	2.54
	SFT_NewData	0.67	2.20	1.80	2.60	3.00	2.48
	PPO	0.67	3.40	2.80	4.40	3.00	3.40
	PPO_NewData	0.65	2.60	2.80	3.60	1.80	2.70
Creative Writing	KoGPT-2	Null	2.20	3.20	3.20	1.80	2.62
	SFT	Null	2.40	2.60	2.80	1.60	2.38
	SFT_NewData	Null	2.20	3.00	2.20	1.60	2.32
	PPO	Null	2.00	1.80	1.80	1.60	1.82
	PPO_NewData	Null	1.80	2.20	2.20	1.40	1.92
Classification & Reasoning	KoGPT-2	0.56	2.40	2.20	1.40	1.60	2.02
	SFT	0.59	1.20	1.40	2.80	1.60	1.78
	SFT_NewData	0.58	2.40	1.20	3.00	2.00	2.14
	PPO	0.58	1.60	1.60	2.20	1.20	1.64
	PPO_NewData	0.60	2.00	2.40	2.60	1.60	2.16
Role-playing & Style Transfer	KoGPT-2	Null	2.40	2.60	3.40	1.80	2.54
	SFT	Null	2.60	3.80	1.80	2.00	2.72
	SFT_NewData	Null	2.80	2.80	3.00	1.80	2.64
	PPO	Null	2.00	3.60	2.40	2.40	2.68
	PPO_NewData	Null	2.80	4.20	4.20	2.60	3.46

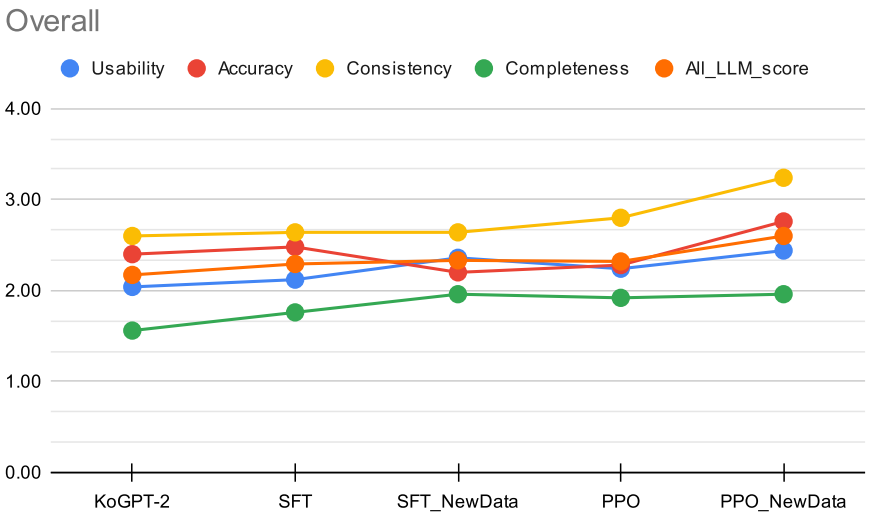


Figure 1: Overall Score Comparison

In the 'Role-playing and Style Transfer' category, the `PPO_NewData` model achieved the highest score of 3.46. This suggests high-quality data is effective for learning complex instructions involving subtle nuances. In 'Simple Q&A', the `PPO_NewData` model also scored highest at 2.76, confirming improvements in basic informational queries. In contrast, 'Creative Writing' and 'Classification and Reasoning' showed complex patterns. The baseline PPO model's performance decreased compared to the SFT model, which could be interpreted as the 'alignment tax'. However, the high-quality dataset mitigated this drop, suggesting that better data can alleviate this phenomenon.

4.5 Analysis of BERTScore F1 Scores

Unlike the LLM-based evaluation, the `bertscore_f1` score did not show significant changes, fluctuating within a narrow range around 0.60. BERTScore measures similarity to a single reference answer. RLHF, however, trains models to generate useful and natural responses reflecting human preferences, not to mimic a single answer, which explains the low correlation. In conclusion, BERTScore is useful for verifying factual consistency but is limited in its ability to holistically measure the overall quality of a model enhanced through RLHF.

4.6 Qualitative Result Examples

We compared responses to the prompt, "Please tell me three major stations on Seoul Subway Line 2's circle line."

- **KoGPT-2 (Base Model):** Failed to grasp the intent and generated irrelevant text.

- **PPO (Baseline Data):** Followed the format but listed incorrect subway lines, showing severe factual errors.
- **PPO_NewData (Enhanced Data):** Mentioned relevant stations, although some were not perfectly accurate. Nevertheless, it generated the most useful response.

These examples show that RLHF with the new dataset improved the model's instruction understanding and response usability.

5 Discussion and Conclusion

This study analyzed the effectiveness of the RLHF methodology for aligning Korean language models from the perspective of dataset quality.

5.1 Summary and Interpretation of Research Findings

The key findings are as follows:

- **Effectiveness of the RLHF Pipeline:** The RLHF pipeline significantly improved instruction-following capabilities compared to the base KoGPT-2 model.
- **Decisive Role of Data Quality:** The quality of the dataset was confirmed to be a critical variable. The model trained with the new dataset consistently outperformed the baseline model.
- **Synergy of High-Quality Data and PPO:** High-quality data played a significant role in improving performance at the PPO stage, implying that a more accurate reward model leads to more effective policy optimization.

5.2 Significance and Limitations of the Study

- **Significance:** This study provides an empirical validation of the importance of data quality within a systematic, stage-by-stage analysis of a Korean RLHF pipeline, offering evidence and direction for future research.
- **Limitations:**
  - **Model Scale:** The KoGPT-2 model is relatively small, so results may not generalize to very large-scale LMs.
  - **Evaluation Data:** The evaluation relied on a small, custom set of 25 prompts.
  - **Experimental Validity:** The experiment was conducted only once. Given the probabilistic nature of LMs, multiple runs are needed to confirm the effects.
  - **Evaluation Method:** The LLM-as-a-judge method is not entirely free from the potential biases of the evaluator LLM itself.

5.3 Conclusion and Future Research Directions

Our conclusion is: "RLHF is an effective method for aligning a Korean language model with user intent, and securing high-quality data is necessary to enhance its effectiveness."

Based on this, future research can consider:

- Applying this methodology to larger-scale Korean base models (e.g., HyperCLOVA, Polyglot-ko).
- Developing a standardized Korean benchmark dataset for more objective evaluation.
- Conducting RLHF experiments with domain-specific datasets to enhance alignment effects.

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A Evaluation Prompts

A.1 Standard Evaluation Prompts (25)

1. **Simple QA (5개)**
   (a) "대한민국의 초대 대통령은 누구인가요?"
   (b) "조선시대 세종대왕이 한글을 창제한 연도는 언제인가요?"
   (c) "컴퓨터의 CPU는 어떤 역할을 하는지 간단히 설명해주세요."
   (d) "물리학에서 뉴턴의 운동 제3법칙(작용-반작용의 법칙)이란 무엇인가요?"
   (e) "서울 지하철 2호선의 주요 순환역 3곳을 알려주세요."
2. **Summarization (5개)**
   (a) "다음 뉴스를 세 문장으로 요약해주세요.
   n'정부의 새로운 부동산 정책이 발표되었다. 이번 정책은 청년 및 신혼부부를 위한 주택 공급 확대에 초점을 맞추고 있으며, 이를 위해 도심 내 유휴부지를

적극 활용할 계획이다. 전문가들은 이번 대책이 단기적인 시장 안정에 기여할 것으로 보면서도, 장기적인 공급 계획이 동반되어야 한다고 조언했다.’"

(b) "아래 제품 설명서의 핵심 기능을 한 문장으로 요약해주세요.
n’새롭게 출시된
’스마트 클리너 A1
’은 인공지능을 탑재하여 집안 구조를 스스로 학습하고, 먼지가 많은 구역을 집중적으로 청소합니다. 또한, 스마트폰 앱과 연동하여 원격으로 청소를 시작하거나 예약할 수 있으며, 배터리가 부족하면 자동으로 충전 스테이션으로 복귀합니다.’"

(c) "다음 글을 바탕으로, 인공지능의 장점과 단점을 각각 한 가지씩 요약해주세요.
n’인공지능은 반복적인 업무를 자동화하여 생산성을 크게 향상시키고, 복잡한 데이터 분석을 통해 인간이 발견하기 어려운 통찰력을 제공한다. 하지만 인공지능의 결정 과정이 불투명한 경우가 많아 책임 소재를 가리기 어렵고, 기존 일자리를 대체할 수 있다는 우려도 존재한다.’"

(d) "다음 영화 줄거리의 전체 내용을 결말을 포함하여 요약해주세요.
n’탐정 김철수는 실종된 과학자를 찾는 의뢰를 받는다. 수사 과정에서 과학자가 시간을 되돌리는 기계를 발명했으며, 거대 조직이 이를 노리고 있다는 사실을 알게 된다. 김철수는 조직의 방해를 뚫고 마침내 과학자를 구하지만, 기계는 이미 망가진 후였다. 그는 세상을 구했지만, 과거를 바꿀 기회는 영원히 잃게 되었다.’"

(e) "아래 역사적 사건에 대한 설명을 읽고, 사건의 원인과 결과를 요약해주세요.
n’19세기 산업혁명은 증기기관의 발명과 같은 기술 혁신으로 시작되었다. 이로 인해 공장제 기계 공업이 발달하며 대량 생산이 가능해졌고, 농업 사회에서 산업 사회로의 급격한 전환이 이루어졌다. 하지만 도시로의 인구 집중, 열악한 노동 환경, 환경 오염과 같은 새로운 사회 문제들이 발생하기도 했다.’"

3. **Creative Writing (5개)**

(a) "‘가을비’를 주제로 짧은 시를 한 편 써주세요."

(b) "우주를 여행하는 고양이가 주인공인 짧은 동화의 시작 부분을 만들어주세요."

(c) "새로 출시될 스마트폰의 광고 문구를 3가지 제안해주세요. 혁신적인 카메라 기능을 강조해야 합니다."

(d) "미래의 서울 모습을 상상하여 한 문단으로 묘사해주세요."

(e) "주인공이 중요한 결정을 앞두고 갈등하는 내용의 짧은 드라마 대본을 작성해주세요."

4. **Classification and Reasoning (5개)**

(a) "다음 영화 리뷰의 감정은 ’긍정’, ’부정’, ’중립’ 중 무엇인가요?:
n’음식은 맛있었지만, 서비스가 너무 느려서 아쉬웠어요.’"

(b) "다음 문장을 읽고 사실과 의견을 구분해주세요.
n’에베레스트 산은 세계에서 가장 높은 산이며, 그 경치는 정말 아름답다.’"

(c) "A는 B보다 키가 크고, B는 C보다 키가 크다. A와 C 중 누가 더 키가 큰가?"

(d) "다음 목록을 생존에 중요한 순서에 따라 재정렬하고 그 이유를 설명해주세요.
n’물, 음식, 공기, 집’"

(e) "다음 글의 핵심 주장을 한 문장으로 파악해주세요.
n’최근 무분별한 일회용품 사용으로 인해 환경 오염이 심각한 수준에 이르렀다. 편리함만을 추구하기보다는 미래 세대를 위해 조금의 불편함을 감수하더라도 다회용품 사용을 생활화하는 사회적 합의가 시급하다.’"

5. **Role-playing and Style Transfer (5개)**
   (a) "당신은 조선시대의 왕입니다. 백성들에게 풍년을 기원하는 교지를 내려주세요."
   (b) "5살 아이도 이해할 수 있도록 양자역학의 기본 개념을 매우 쉽게 설명해주세요."
   (c) "SF 소설의 인공지능 캐릭터처럼 감정이 배제된 채, 오늘 날씨를 보고해주세요."
   (d) "친한 친구에게 말하듯이, 오늘 하루 있었던 일을 편하게 이야기해주세요."
   (e) "다음 문장을 전문가가 쓴 것처럼 전문적인 용어를 사용하여 더 어렵게 바꿔주세요.
   n'컴퓨터를 끄기 전에 작업한 내용을 저장하는 것이 좋습니다.'"

A.2 Reference Answers (Labels) (13)

0: "이승만"

1: "1443년"

2: "컴퓨터의 중앙 처리 장치로, 모든 데이터 처리와 연산을 수행하며 시스템 전체를 제어하는 핵심 부품이다."

3: "모든 작용에는 크기가 같고 방향이 반대인 반작용이 항상 존재한다는 법칙이다."

4: "강남역, 잠실역, 신도림역"

5: "정부가 청년과 신혼부부를 위한 주택 공급 확대 정책을 발표했다. 도심 내 유휴 부지를 활용하는 것이 주요 방안이며, 전문가들은 장기적 공급 계획의 필요성을 강조했다."

6: "'스마트 클리너 A1'은 AI를 통해 집안 구조를 학습하여 효율적으로 청소하고, 스마트폰 앱으로 원격 제어가 가능한 자동 로봇 청소기다."

7: "인공지능의 장점은 반복 업무 자동화를 통한 생산성 향상이며, 단점은 결정 과정의 불투명성으로 인한 책임 소재의 문제이다."

8: "탐정 김철수는 실종된 과학자를 찾는 과정에서 시간 여행 기계를 둘러싼 거대 조직의 음모를 파헤치고, 결국 세상을 구하지만 과거를 바꿀 기회는 잃게 된다."

9: "산업혁명의 원인은 증기기관 발명과 같은 기술 혁신이며, 그 결과 대량 생산이 가능해져 산업 사회로 전환되었지만 도시 문제와 환경 오염 같은 부작용도 발생했다."

15: "중립"

16: "사실: 에베레스트 산은 세계에서 가장 높은 산이다. 의견: 그 경치는 정말 아름답다."

17: "A"

B LLM Score-Cali Ration Criteria

You are an expert evaluator assessing the responses of a language model. Score the given 'model's response' according to the criteria below and provide a score for each item.

Evaluation Criteria:

1. **Usefulness**: Does the response clearly understand the user's intent in the 'original prompt' and provide core, useful information accordingly? (1-10 points)

2. **Accuracy**: Is the response free of factual errors and logically sound? (1-10 points)
3. **Coherence**: Are the sentences natural, and is the overall flow of the text logically consistent? (1-10 points)
4. **Completeness**: Does the response contain sufficient information, making it complete enough that the user would not need to ask follow-up questions? (1-10 points)

C Task-specific figure

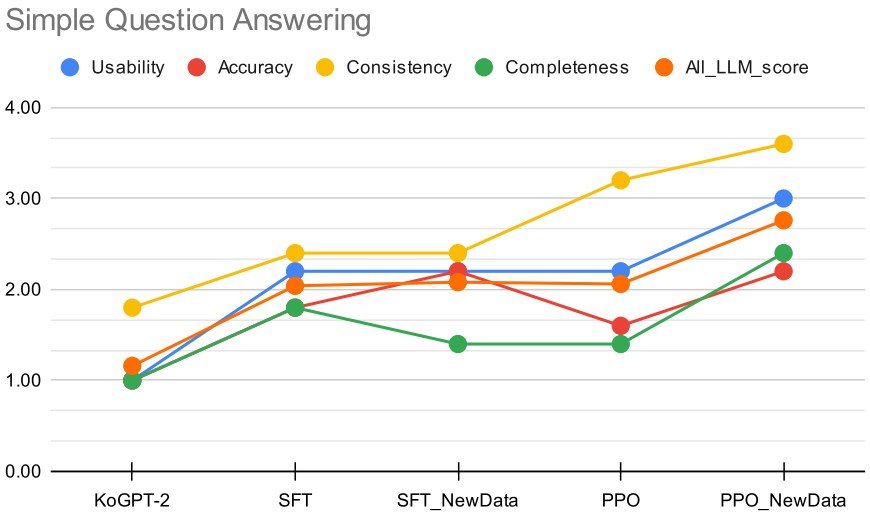


Figure 2: Overall Score Comparison

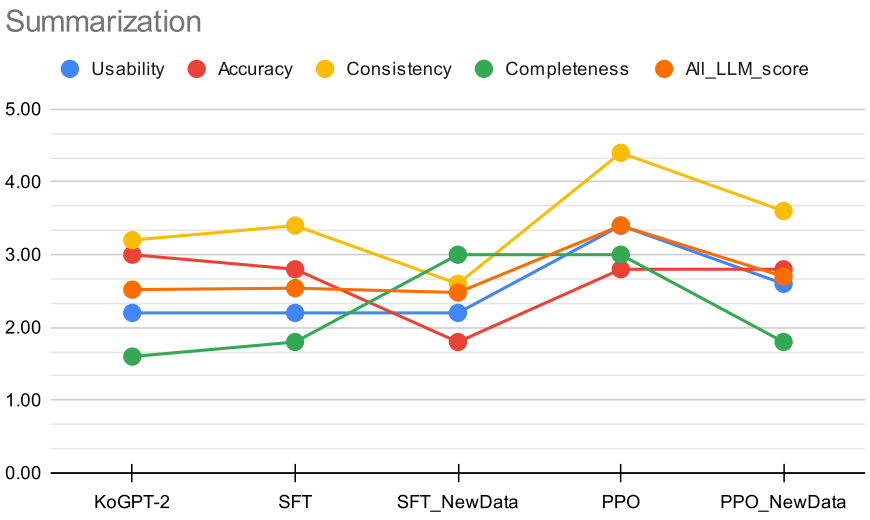


Figure 3: Overall Score Comparison

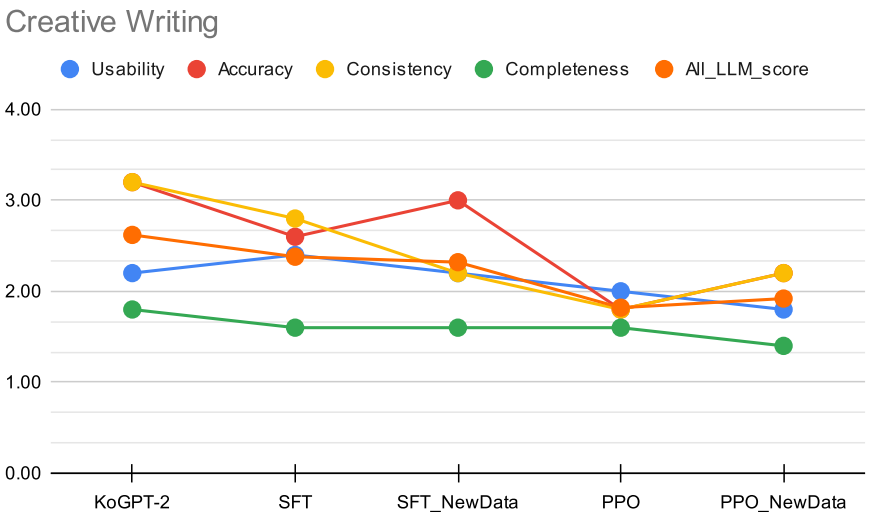


Figure 4: Overall Score Comparison

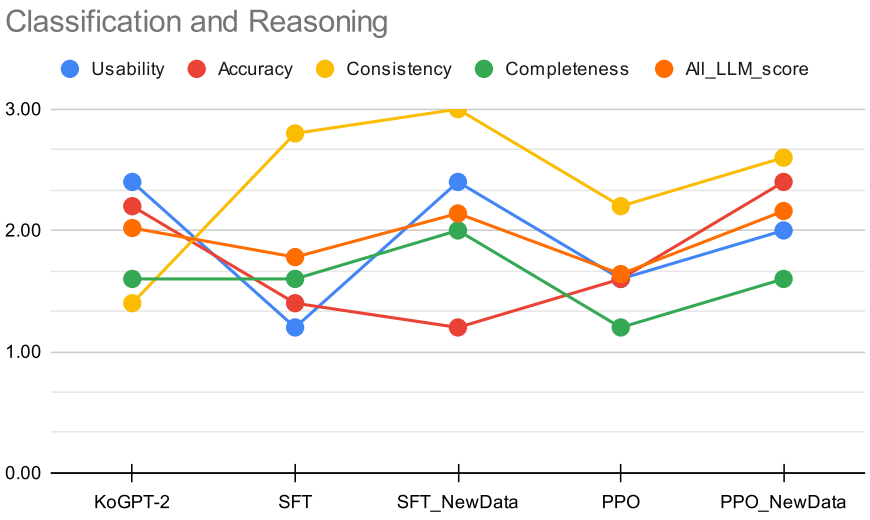


Figure 5: Overall Score Comparison

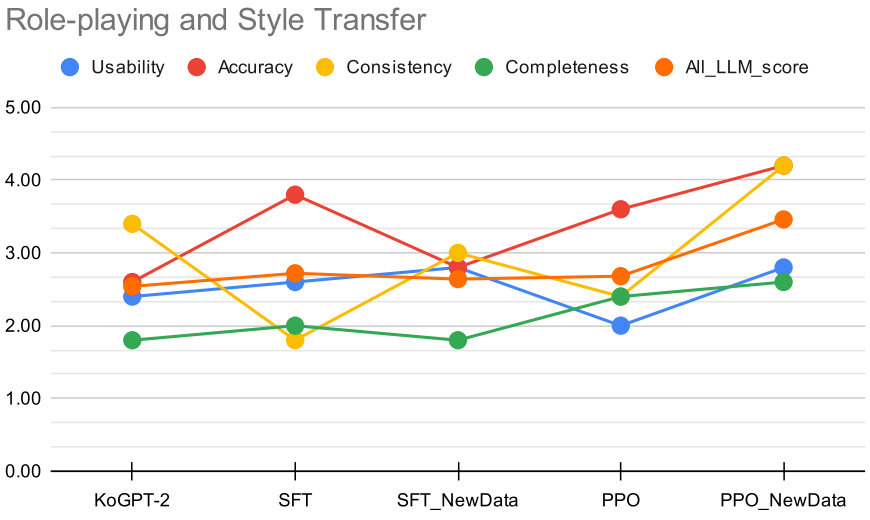


Figure 6: Overall Score Comparison